

IEEE Transactions on Intelligent Transportation Systems (ITS) Special Issue on IEEE Workshop on Connected and Autonomous Vehicles, October 27–28, 2022

THE IEEE Workshop on Connected and Autonomous Vehicles (CAV) was organized to bring experts from intelligent transportation systems (ITS) in this specific area together to discuss some of the latest findings and research results, applicable technologies, issues and limitations, standards development activities, societal impacts, and future research and development directions. This workshop was sponsored by the IEEE Intelligent Transportation Systems (ITS) Society, with the support of the Mechanical Engineering Department of Virginia Tech. Dr. Azim Eskandarian, Nicholas, and Rebecca Des Champs Chair and the Head of the Mechanical Engineering Department at Virginia Tech at the time of this workshop (and currently the Dean of Engineering and the Alice T. and William H. Goodwin Jr. Endowed Chair at Virginia Commonwealth University) chaired this workshop. Dr. Monastir Abbas, Professor of Civil Engineering at Virginia Tech, was the Workshop's Co-Chair. They invited 19¹ renowned experts, each specializing in specific areas of CAV. They were members of academia, industry, government, and related professional organizations, including IEEE, ITE, and SAE. The workshop was held on October 27–28, 2022, at Hyatt Hotel, 7901 Tysons One Place, Tysons Corner, VA 22102 USA.

The workshop provided an open forum for discussions and conversations of these experts about important issues, breakthroughs, and advanced landmark research projects highlighting significant challenges in this field rather than focusing on specific research projects with narrower scopes, typically discussed in other appropriate venues like conference proceedings. Therefore, this compendium does not include regular research, survey, or opinion papers. However, it consists of the material presented by each presenter, including a title, abstract, presentation slides, and the associated discussion summary notes.

The workshop created a focused venue for experts' consultation on connected autonomous vehicles, their interactions with infrastructure and road users, and their impacts. It emphasized future challenges, shortcomings, and possible solutions. Experts expressed their opinions and the latest findings objectively and unbiasedly to provide valuable guidance for researchers, government managers, and standard-developing organizations. While the intent was not

to discuss specific standards, the guidance and advice from this forum could help define high-level requirements for future standards. The presentations in this volume highlight the critical issues to consider and alert to potential shortfalls as the commercial community rushes to develop and introduce these automated and connected autonomous systems to the market.

This unique “meeting of the minds” workshop brought together researchers from the ITS Society, cognizant government agencies (e.g., the Department of Transportation, the Department of Energy, and the National Science Foundation), and other experts leading efforts related to automated and autonomous transportation. Although the field is too broad to be captured comprehensively in any one workshop or conference, the results are still helpful to many interested in these topics. The workshop contents can help as a roadmap on how IEEE and other stakeholder organizations can position themselves effectively in the transportation revolution of automated and autonomous vehicles. It includes advice on forthcoming “standards” development activities, sample relevant technologies, and new opportunities.

Since we did not record the workshop due to logistical and other related issues and concerns, the organizers and participants decided that its contents were too precious not to make available to the public, especially readers in this field. Therefore, the best way to preserve this timely and exciting dialogue was through preservation in a published volume by the IEEE. As such, we published it in this Special Issue of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, the flagship transactions of the ITS Society. Although this is not the norm for the Special Issues of this transaction, it allows the readers to get a first-hand glimpse of this landmark workshop's topics, presentations, and discussions. The authors, all presenters, and participants hope that the readers find the contents helpful in their quest for new research and projects in ITS related to CAVs.

AZIM ESKANDARIAN, *Editor-in-Chief*
Dean, College of Engineering
and Alice T. and William H. Goodwin Jr. Endowed Chair
Virginia Commonwealth University
Richmond, VA 23284 USA

MONTASIR ABBAS
Department of Civil and Environmental Engineering
Virginia Tech
Blacksburg, VA 24061 USA

¹This Special Issue contains 18 out of 19 presentations and discussions; one subject was unavailable due to logistical restrictions.